

BS849 Team5 Final Project

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Introduction

Boston University conducted two studies assessing longevity and healthy aging among the offspring of centenarians and a control group. Our dataset (tics.data.2021.csv) focuses on the outcome Telephone Interview for Cognitive Status (TICS01–TICS05), a longitudinal measure of cognitive health. The dataset includes continuous predictors such as BMI, years of education, and age, as well as categorical predictors including hypertension, diabetes, smoking status, aspirin use, stroke, coronary artery disease, cancer, and heart attack.

Our research questions are:

1. Do the offspring of centenarians exhibit different cognitive health trajectories (TICS scores) over the five timepoints compared to the control group?
2. Are there potential confounders that strengthen or weaken the association between TICS scores and the groups?
3. Does the baseline TICS score differ between groups after accounting for potential confounders?
4. Are there subgroups that show differences in TICS scores by group? How should missing data in the dataset be handled, and does it meaningfully impact our findings?

Overview

Our dataset has 27 variables and 762 observations. We then conduct univariate analysis on each variable at a time to help us determine which predictors might be potential confounders. For the continuous variables, we will quickly check their distribution before determining whether we'll analyze them with a t-test or a Wilcoxon test.

Distribution

All variables follow normal distribution (determined by a bell-shaped histogram) except for BMI, which appears right skewed, therefore we conducted a Wilcoxon Test on BMI. The downloaded binary packages are in /var/folders/bg/jgtlht4d6cvfn0vy28bp75k40000gn/T//RtmpKwdysG/downloaded_packages

```
knitr::include_graphics("./Plot/Q1_Table1.png")
```

Main Findings from Part 1

The comparison between the control group (N= 270) and the centenarian offspring (N=492) shed light on several baseline differences that are statistically significant. The offspring group has a slightly lower BMI (26.2 v 27.0, $p = 0.017$) and fewer individuals in the offspring group have ever smoked (53% v 64%, $p=0.003$). There is a lower prevalence in stroke among the centenarian offspring (1.6% v 4.4%), as well as lower rates of diabetes (5.1% v 12%, $p = <0.001$), hypertension (30% v 45%, $p = <0.001$), and coronary artery disease (3.7% v 10.0%, $p=<0.001$). Lastly, there was a statistically significant difference in sex distribution, with more females in the control group (37% v 52%), and more males in the offspring group (63% v 48%, $p = <0.001$). All other factors were not found to be statistically significant.

Characteristic	Control N = 270 ¹	Centennium Offspring N = 492 ¹	p-value ²
BMI	Known 27.0 (24.4, 30.1) Unknown 21	26.2 (23.6, 29.6) 17	0.017
Age01	75 (7)	75 (7)	0.3
sex	0 140 (52%) 1 130 (48%)	181 (37%) 311 (63%)	<0.001
SH.Ever.Smoked.	No 96 (36%) Yes 174 (64%)	232 (47%) 260 (53%)	0.003
MC.Aspirin	No 131 (49%) Yes 139 (51%) Unknown 0	275 (56%) 216 (44%) 1	0.057
MC.Stroke	No 258 (96%) Yes 12 (4.4%)	484 (98%) 8 (1.6%)	0.037
MC.Diabetes.Mellitus	No 237 (88%) Yes 33 (12%)	467 (95%) 25 (5.1%)	<0.001
MC.HTN	No 149 (55%) Yes 121 (45%)	345 (70%) 147 (30%)	<0.001
MC.Coronary.Artery.Disease	No 243 (90%) Yes 27 (10%)	474 (96%) 18 (3.7%)	<0.001
MC.Cancer	No 197 (73%) Yes 73 (27%)	373 (76%) 119 (24%)	0.4
MC.Heart.Attack	No 253 (94%) Yes 17 (6.3%)	471 (96%) 21 (4.3%)	0.3
Years.of.Education	15.65 (2.86)	15.87 (2.93)	0.3
TICS01	13 (11, 16) Unknown 1	14 (11, 16) 1	0.2
¹ Median (Q1, Q3); Mean (SD); n (%)			
² Wilcoxon rank sum test; Welch Two Sample t-test; Pearson's Chi-squared test			

Figure 1: Distribution table of covariates

Distributions of TICS score at baseline between offspring of centenarians and controls after adjusting for possible confounders

Purpose of question 2 is to test whether TICS01 at baseline differs by group (ptype), after adjusting for confounders identified in Table 1.

After adjusting for age, sex, BMI, smoking, stroke, diabetes, hypertension, and CAD, baseline TICS scores did not differ significantly between centenarian offspring and controls: $\beta_1 = 0.31$, 95% Credible Interval: -0.29 to 0.91. The 95% credible interval includes zero, providing no evidence of a group difference in cognitive function at baseline. All three MCMC chains converged satisfactorily for the baseline model. The Gelman-Rubin PSRF was $R \leq 1.05$ for all parameters (multivariate PSRF = 1.04), and all Geweke Z-scores were within ± 2 .

As expected, older age was associated with lower TICS scores ($\beta_2 = -0.21$, 95% CI: -0.24 to -0.17), and male sex was associated with higher scores ($\beta_3 = 1.40$, 95% CI: 0.80 to 1.87). The remaining covariates (BMI, smoking, stroke, diabetes, hypertension, CAD) did not show credible independent effects on baseline TICS after mutual adjustment, as their 95% credible intervals all included zero. The residual standard deviation was estimated at $\sigma = 3.87$ (95% CI: 3.68 to 4.07), reflecting substantial between-person variability.

```
knitr::include_graphics("./Plot/Q2_Summary.png")
```

```
> summary(samples.base)
```

Iterations = 10001:20000
Thinning interval = 1
Number of chains = 3
Sample size per chain = 10000

1. Empirical mean and standard deviation for each variable,
plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE	
beta0	29.79252	1.65771	0.0095708	0.1469735	beta0
beta1	0.31333	0.30621	0.0017679	0.0039795	beta1# CenOff vs Control
beta2	-0.20592	0.01911	0.0001103	0.0015466	beta2 # age at baseline
beta3	1.39196	0.29640	0.0017112	0.0036109	beta3 # sex
beta4	-0.05254	0.03206	0.0001851	0.0015833	beta4 # BMI
beta5	0.19623	0.28871	0.0016668	0.0032867	beta5 # ever smoked
beta6	-0.81988	0.88780	0.0051257	0.0053778	beta6 # stroke history
beta7	-0.64867	0.55314	0.0031935	0.0043472	beta7 # diabetes
beta8	-0.11909	0.30942	0.0017864	0.0032362	beta8 # hypertension
beta9	-0.55441	0.62540	0.0036108	0.0051136	beta9 # CAD
sigma	3.86973	0.10027	0.0005789	0.0005976	

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
beta0	26.6624	28.626062	29.79330	30.93450	33.10843
beta1	-0.2881	0.107368	0.31143	0.51595	0.91657
beta2	-0.2424	-0.218818	-0.20579	-0.19282	-0.16869
beta3	0.8054	1.193753	1.39584	1.58964	1.96917
beta4	-0.1158	-0.074542	-0.05189	-0.03017	0.00922
beta5	-0.3668	0.001209	0.19677	0.38863	0.75939
beta6	-2.5622	-1.416690	-0.81869	-0.21851	0.92543
beta7	-1.7356	-1.021588	-0.64406	-0.27665	0.43819
beta8	-0.7257	-0.329009	-0.11973	0.08879	0.49132
beta9	-1.7762	-0.976375	-0.55473	-0.12843	0.66278
sigma	3.6803	3.800085	3.86809	3.93664	4.07096

Figure 2: Summary of TICS baseline between offspring of centenarians and controls after adjusting for possible confounders

```
knitr::include_graphics("./Plot/Q2_Diagnosis.png")
```

```
knitr::include_graphics("./Plot/Q2_gelman-rubin.png")
```

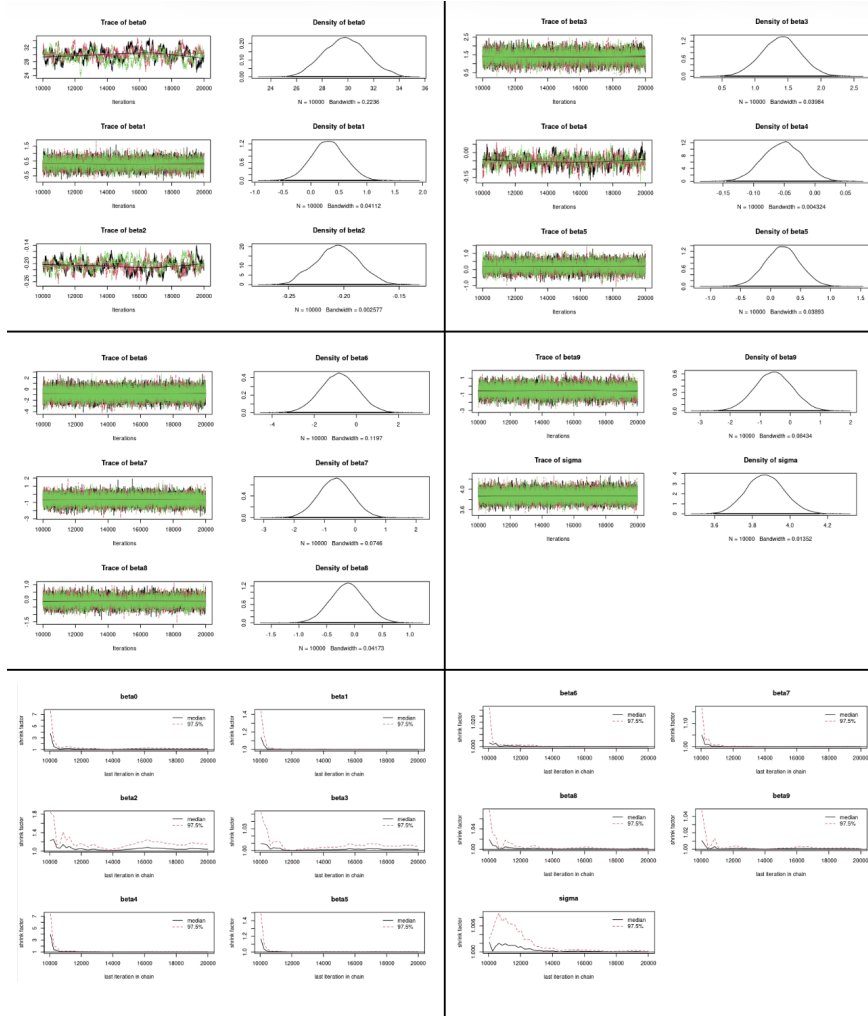


Figure 3: Diagnosis plot for TICS baseline model

```

[[1]]
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

    beta0    beta1    beta2    beta3    beta4    beta5    beta6    beta7    beta8    beta9    sigma
-1.33457 -0.90063  1.61327  1.49704 -0.01645  0.44286 -0.70503  0.45868 -1.61526 -1.91489  1.93052

[[2]]
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

    beta0    beta1    beta2    beta3    beta4    beta5    beta6    beta7    beta8    beta9    sigma
-0.06678  0.37516  0.36628 -0.94678 -2.06471 -1.06865  0.95840  1.87299  2.41428 -1.76319 -0.36216

[[3]]
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

    beta0    beta1    beta2    beta3    beta4    beta5    beta6    beta7    beta8    beta9    sigma
-1.0336  1.0103  1.0595 -0.3292  0.1832 -1.5185 -0.6145  0.6170  0.2639 -2.1684 -0.5887

> gelman.diag(samples.base)
Potential scale reduction factors:

      Point est. Upper C.I.
beta0      1.01      1.02
beta1      1.00      1.00
beta2      1.01      1.01
beta3      1.00      1.00
beta4      1.01      1.01
beta5      1.00      1.00
beta6      1.00      1.00
beta7      1.00      1.00
beta8      1.00      1.00
beta9      1.00      1.00
sigma      1.00      1.00

Multivariate psrf

1.01

```

Figure 4: Gelman-Rubin statistics of the baseline model

Hierarchical models to investigate differences in the rate of change of TICS between offspring of centenarians and controls

In the hierarchical model of TICS change at follow ups 2–5, the estimated mean slope in the control group was $\beta = -0.31$ TICS points per average follow up interval (posterior mean -0.30596, 95% credible interval -0.39552, -0.21717), indicating clear cognitive decline over time. The group by time interaction δ , representing the difference in slope between centenarian offspring and controls, had posterior mean 0.14 with a 95% credible interval 0.0539 to 0.2271, suggesting that offspring experienced a modestly slower rate of decline than controls. Thus, the adjusted offspring slope $\beta + \delta$ was approximately -0.17 points per follow up interval, conditional on baseline TICS01, baseline age, BMI, sex, smoking, hypertension, diabetes, stroke, and coronary artery disease. Baseline TICS01 strongly predicted subsequent TICS levels (gamma about 0.48, 95% credible interval 0.42, 0.53). Older baseline age and higher BMI were associated with lower TICS scores over follow up, male sex was associated with higher TICS, and prior stroke was associated with lower TICS trajectories, while hypertension and smoking showed positive associations with TICS after adjustment.

```
knitr::include_graphics("./Plot/q3_summary.png")
```

Gelman–Rubin statistics were 1.00–1.01 for all parameters, and trace, density, and autocorrelation plots for β and δ showed well mixed chains and steadily decaying autocorrelation, indicating satisfactory convergence and large effective sample sizes for the key slope parameters, whereas γ exhibited slower autocorrelation decay but acceptable convergence overall.

```
knitr::include_graphics("./Plot/q3_gelman-rubin.png")
```

```
knitr::include_graphics("./Plot/q3_trace_density.png")
```

```
knitr::include_graphics("./Plot/q3_acf.png")
```

Subgroups of study participants with significantly different rates of change of TICS

The Bayesian latent trajectory model identified two distinct groups of participants with different longitudinal patterns of TICS scores.

```
knitr::include_graphics("./Plot/Q4_Summary.png")
```

Class proportions: The estimated group proportions were approximately 45% for Class 1 and 55% for Class 2, indicating a relatively balanced distribution of participants across the two latent trajectory groups.

Baseline differences: The two groups showed similar baseline levels: - Class 1: b_0 is approximately 7.93. - Class 2: b_0 is approximately 8.12.

The credible intervals for these intercepts largely overlap, suggesting that there is no strong evidence of substantial differences in baseline cognitive levels between the two groups.

Rates of change: The two groups exhibited clearly different slopes over time: - Class 1: b_1 is approximately 0.04 (95% CrI includes 0) - Class 2: b_1 is approximately -0.36 (95% CrI entirely below 0)

This indicates that Class 1 shows relatively stable cognitive performance over time, while Class 2 demonstrates a significant decline in TICS scores. The negative slope for Class 2 is both larger in magnitude and statistically meaningful, suggesting meaningful heterogeneity in longitudinal change.

Effect of baseline cognition: Baseline TICS (TICS01) was positively associated with future TICS scores in both groups: - Class 1: g_0 is approximately 0.58 - Class 2: g_0 is approximately -0.43

This indicates that individuals with higher baseline cognition tend to maintain higher cognitive performance over time. The effect is slightly stronger in Class 1.

Cluster separation:

```
knitr::include_graphics("./Plot/Q4_ClassSeperation.png")
```

1. Empirical mean and standard deviation for each variable, plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
b_age	-0.17853	0.01617	4.669e-05	0.0001462
b_bmi	-0.07358	0.02272	6.559e-05	0.0001539
b_cad	0.54474	0.48831	1.410e-03	0.0033483
b_diab	-0.61693	0.43571	1.258e-03	0.0029845
b_htn	0.33458	0.23207	6.699e-04	0.0019673
b_sex	1.17196	0.22420	6.472e-04	0.0026094
b_smoke	0.30408	0.21466	6.197e-04	0.0022404
b_stroke	-1.60072	0.74233	2.143e-03	0.0047457
beta	-0.30539	0.04517	1.304e-04	0.0005875
delta	0.14002	0.04415	1.275e-04	0.0005105
gamma	0.47646	0.02755	7.953e-05	0.0007649
mu_alpha	6.69158	0.45312	1.308e-03	0.0162145
sigma	2.77176	0.05191	1.498e-04	0.0002268
sigma_alpha	2.14081	0.09574	2.764e-04	0.0005841

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
b_age	-0.21010	-0.18950	-0.17855	-0.16772	-0.14670
b_bmi	-0.11772	-0.08895	-0.07362	-0.05826	-0.02899
b_cad	-0.41544	0.21455	0.54330	0.87572	1.50096
b_diab	-1.47184	-0.91134	-0.61491	-0.32264	0.23379
b_htn	-0.12122	0.17860	0.33444	0.49037	0.79058
b_sex	0.72932	1.02210	1.17359	1.32345	1.60959
b_smoke	-0.11881	0.15958	0.30479	0.44862	0.72361
b_stroke	-3.05361	-2.10221	-1.60141	-1.10123	-0.14174
beta	-0.39464	-0.33571	-0.30517	-0.27488	-0.21714
delta	0.05377	0.11007	0.14006	0.16989	0.22634
gamma	0.42250	0.45784	0.47649	0.49508	0.53045
mu_alpha	5.80285	6.38675	6.69120	6.99598	7.57225
sigma	2.67217	2.73652	2.77082	2.80633	2.87617
sigma_alpha	1.95596	2.07586	2.13967	2.20485	2.32988

Figure 5: Summary statistics of the heirarchical model

Potential scale reduction factors:

	Point est.	Upper C.I.
b_age	1.00	1.00
b_bmi	1.00	1.00
b_cad	1.00	1.00
b_diab	1.00	1.00
b_htn	1.00	1.00
b_sex	1.00	1.00
b_smoke	1.00	1.00
b_stroke	1.00	1.00
beta	1.00	1.00
delta	1.00	1.00
gamma	1.01	1.02
mu_alpha	1.01	1.03
sigma	1.00	1.00
sigma_alpha	1.00	1.00

Figure 6: Gelman-Rubin convergence statistics of the hierarchical model

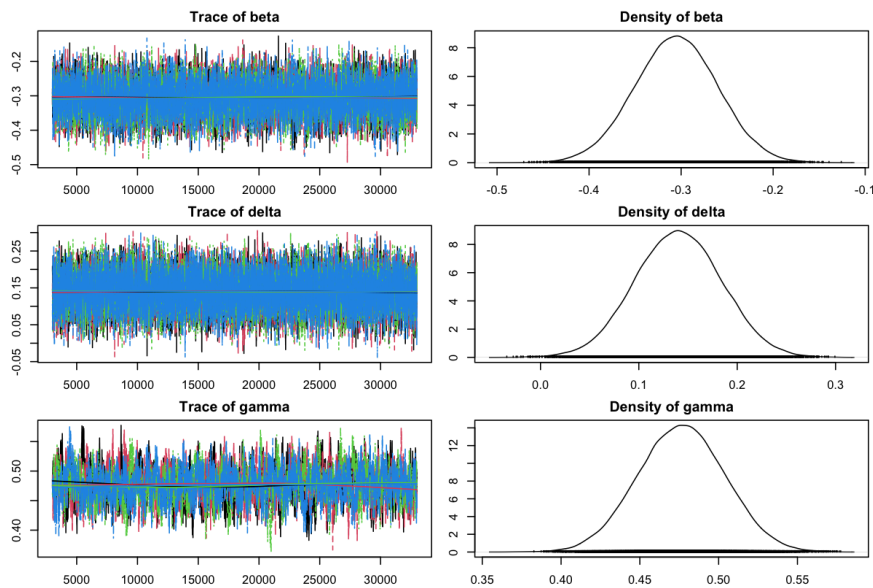


Figure 7: Trace and density plot of the hierarchical model

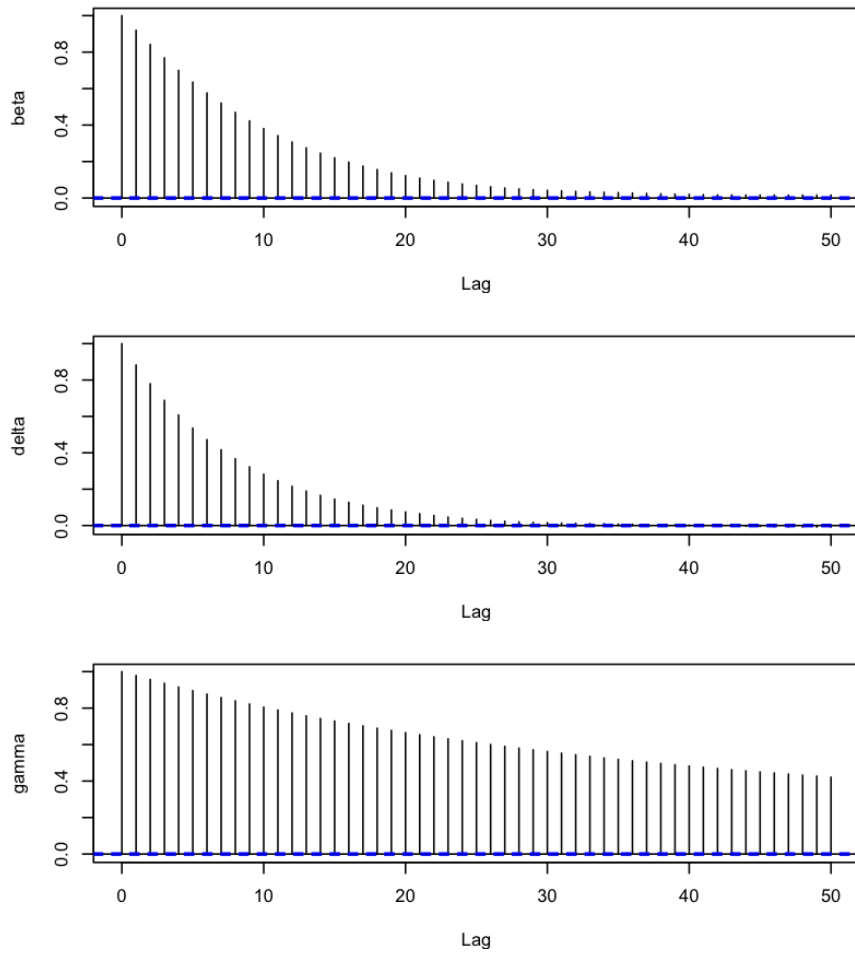


Figure 8: Autocorrelation plot of the β , δ , and γ values of the hierarchical model

```
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
```

```
##
##      Mean      SD Naive SE Time-series SE
## b0[1]  7.93357 0.597460 3.449e-03    3.443e-02
## b0[2]  8.11943 0.590495 3.409e-03    3.026e-02
## b1[1]  0.04439 0.053309 3.078e-04    1.061e-03
## b1[2] -0.35703 0.047480 2.741e-04    9.420e-04
## g0[1]  0.57584 0.041286 2.384e-04    2.074e-03
## g0[2]  0.42598 0.039631 2.288e-04    1.804e-03
## theta[1] 0.44858 0.041835 2.415e-04    6.044e-04
## theta[2] 0.55142 0.041835 2.415e-04    6.044e-04
## tau    0.11608 0.005014 2.895e-05    3.348e-05
```

```
## 2. Quantiles for each variable:
```

```
##
##      2.5%      25%      50%      75%      97.5%
## b0[1]  6.66684  7.567984  7.96422  8.33297  9.0484
## b0[2]  6.89446  7.750202  8.14294  8.51294  9.2276
## b1[1] -0.05856  0.008398  0.04437  0.07978  0.1507
## b1[2] -0.45223 -0.388563 -0.35661 -0.32476 -0.2659
## g0[1]  0.49883  0.548784  0.57331  0.60000  0.6661
## g0[2]  0.35232  0.399458  0.42435  0.45128  0.5082
## theta[1] 0.36657  0.420198  0.44851  0.47670  0.5308
## theta[2] 0.46920  0.523301  0.55149  0.57980  0.6334
## tau    0.10647  0.112598  0.11602  0.11948  0.1260
```

Figure 9: Summary statistics of the subgroups model

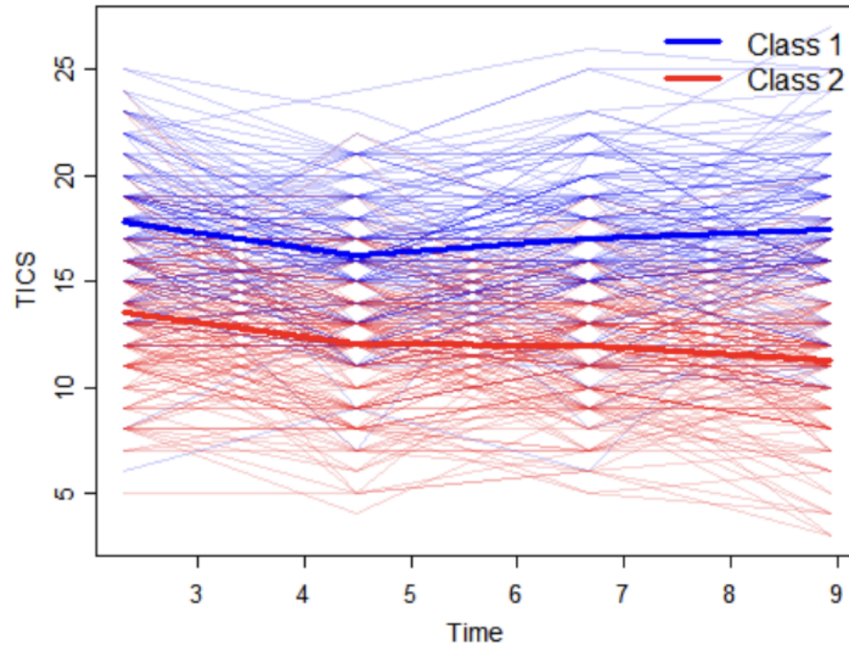


Figure 10: Results of the class separation of the subgroups model

Posterior membership probabilities show clear separation between groups, with most individuals having high probabilities (close to 0 or 1) of belonging to a specific class. This suggests that the model provides meaningful classification.

Trajectory visualization: The trajectory plot shows that the two groups follow distinct patterns: - Class 1 remains relatively stable over time - Class 2 shows a clear downward trend

These patterns support the presence of heterogeneity in cognitive decline trajectories.

Convergence diagnostics:

```
knitr::include_graphics("./Plot/Q4_Diagnosis.png")
```

```
geweke.diag(samp[, params_conv1], frac1 = 0.1, frac2 = 0.5)
## [[1]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
## b0[1] b0[2] b1[1] b1[2]
## -1.0012 -1.0484 2.5722 -0.6823
##
## [[2]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
## b0[1] b0[2] b1[1] b1[2]
## -0.6446 -0.6971 0.6801 1.7443
##
## [[3]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
## b0[1] b0[2] b1[1] b1[2]
## -0.04130 0.07498 -1.30848 -0.56059
plot(samp[, c("b0[1]", "b0[2]", "b1[1]", "b1[2]", "theta[1]", "theta[2]")])
```

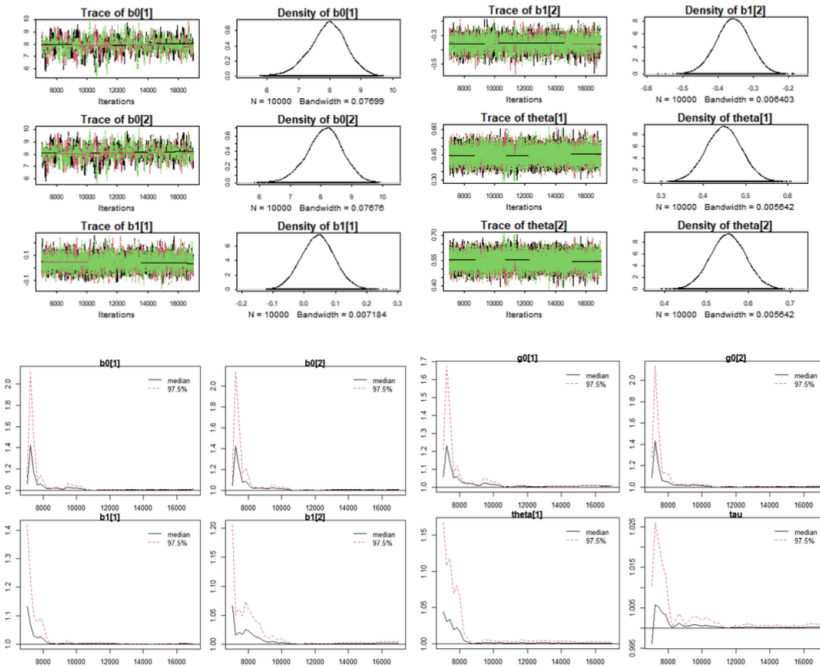


Figure 11: Diagnosis plot for subgroup model

Convergence diagnostics indicate that the model has converged adequately. Gelman-Rubin statistics for all monitored parameters were approximately equal to 1, and the Gelman-Rubin diagnostic plots show that all parameters rapidly converge to values close to 1 and remain stable, indicating good convergence across chains. Geweke diagnostics were generally within acceptable ranges, and trace plots showed good mixing behavior.

How robust results are to the missing data mechanism

To evaluate the robustness of the results found in part 3, we decided to impute the data and rerun the model. A preliminary analysis of how many missing data entries there are across TICS show an increase in percentage. The first TICS had only 0.3% missing data for all 762 individuals, and by the last TICS, the amount of missing data was up to 59.2% as reported in the table below.

```
knitr::include_graphics("./Plot/Q5_na_summary.png")
```

TICS01	TICS02	TICS03	TICS04	TICS05
0.3	9.7	16.4	30.2	59.2

Figure 12: Summary of NA values in each TICS in percentage

An additional analysis using a logistic regression model was used to determine if our significant cofounders were associated with missing values. Within this test, if any of the variables has a significant value, then it would be conformation the data is not missing completely at random. TICS05 was used here as this was determined to have the most NA values present, and in context makes the most sense as by this point in the study, the participants are more likely to be experiencing cognitive decline.

```
knitr::include_graphics("./Plot/Q5_model_mar_summary.png")
```

```
Call:
glm(formula = miss_TICS05 ~ ptype + TICS01 + Age01 + sex + BMI +
    SH.Ever.Smoked. + MC.Stroke + MC.Diabetes.Mellitus + MC.HTN +
    MC.Coronary.Artery.Disease, family = binomial, data = TICS)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)      -0.233469   1.140854  -0.205 0.837850
ptype             -0.577980   0.176388  -3.277 0.001050 **
TICS01            -0.072201   0.020789  -3.473 0.000515 ***
Age01              0.026648   0.012029   2.215 0.026736 *
sex               -0.334919   0.167058  -2.005 0.044983 *
BMI               -0.002322   0.017036  -0.136 0.891572
SH.Ever.Smoked.    0.301520   0.160523   1.878 0.060332 .
MC.Stroke          0.205575   0.549155   0.374 0.708146
MC.Diabetes.Mellitus -0.032066   0.325050  -0.099 0.921416
MC.HTN             0.043824   0.173547   0.253 0.800639
MC.Coronary.Artery.Disease 0.173732   0.381472   0.455 0.648803
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 984.08  on 721  degrees of freedom
Residual deviance: 923.11  on 711  degrees of freedom
(40 observations deleted due to missingness)
AIC: 945.11

Number of Fisher Scoring iterations: 4
```

Figure 13: Summary of the missing data model estimating the MAR

The model resulted in significant association with ptype, baseline, age01, and sex. After these results, the TICS data was imputed for TICS01-TICS05, age01, and bmi, and the hierarchical model was ran again with the new imputed data rather than the complete case analysis. Convergence of the new data was good with a Gelman diagnostic of 1 for each covariate.

```
knitr::include_graphics("./Plot/Q5_test_hier_imp_summary.png")
```

```
Iterations = 5001:55000
Thinning interval = 1
Number of chains = 3
Sample size per chain = 50000
```

1. Empirical mean and standard deviation for each variable, plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
beta_0	-0.04824	0.02618	6.761e-05	0.0002779
beta_1	0.29606	0.02063	5.327e-05	0.0004873
beta_2	0.87720	0.17562	4.534e-04	0.0016644
beta_3	-0.09531	0.01223	3.158e-05	0.0001120
beta_4	-0.05854	0.01823	4.707e-05	0.0001031
beta_5	0.20588	0.16804	4.339e-04	0.0015452
beta_6	-0.66082	0.51837	1.338e-03	0.0028564
beta_7	-0.04651	0.32233	8.322e-04	0.0018859
beta_8	0.16993	0.17948	4.634e-04	0.0013017
beta_9	0.19556	0.36476	9.418e-04	0.0021556
delta	0.02872	0.02570	6.635e-05	0.0002257
mu.alpha	9.24941	0.34380	8.877e-04	0.0103298
sigma	2.78812	0.04138	1.068e-04	0.0001500
sigma.alpha	1.78081	0.07638	1.972e-04	0.0004312

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
beta_0	-0.09978	-0.06602	-0.04811	-0.03043	0.002645
beta_1	0.25539	0.28216	0.29620	0.30986	0.336536
beta_2	0.52987	0.75931	0.87792	0.99662	1.219123
beta_3	-0.11930	-0.10356	-0.09532	-0.08711	-0.071223
beta_4	-0.09436	-0.07085	-0.05851	-0.04630	-0.022764
beta_5	-0.12480	0.09221	0.20629	0.31954	0.533708
beta_6	-1.67072	-1.00952	-0.66156	-0.31176	0.354706
beta_7	-0.67841	-0.26394	-0.04684	0.17104	0.585649
beta_8	-0.18217	0.04853	0.16952	0.29170	0.520576
beta_9	-0.52212	-0.05014	0.19481	0.44091	0.910973
delta	-0.02184	0.01137	0.02881	0.04602	0.078919
mu.alpha	8.57143	9.01816	9.25039	9.47897	9.927768
sigma	2.70858	2.75992	2.78757	2.81561	2.870486
sigma.alpha	1.63244	1.72908	1.78035	1.83169	1.931974

Figure 14: Summary of the hierarchical model imputating the missing values with the mean at each visit

```
knitr::include_graphics("./Plot/Q5_gelman_diag.png")
```

In comparison to the complete case analysis, most of the covariates that were significant remained significant. This includes baseline score, sex, age1, and BMI, with borderline significance for the rate of change, and the group interaction effect, at a confidence interval of (-0.09978, 0.002645) and (-0.02184, 0.078919), respectively. This is different from the complete case analysis where the intervals were not borderline and crossed 0 with stronger magnitude.

It is also of note that the magnitude of the mean scores were lower than with the complete case analysis across all variables. For example, in the complete case analysis, the term delta had a mean of 0.14, but when the data is imputed this mean decreases to 0.02872. In another example, the mean for the rate of change was -0.31 previously, and is not -0.04. This result can be attributed to the fact that the missing entries were changed to represent the average participant, reducing the amount of variance within the data, and reducing the complexity of the relationships. Overall, the results are not robust to mean imputation of the TICS variables, and the way the missing data is handled cannot be by replacing them with average values, as this masks the true relationship of these covariates.

Potential scale reduction factors:

	Point est.	Upper C.I.
beta_0	1	1
beta_1	1	1
beta_2	1	1
beta_3	1	1
beta_4	1	1
beta_5	1	1
beta_6	1	1
beta_7	1	1
beta_8	1	1
beta_9	1	1
delta	1	1
mu.alpha	1	1
sigma	1	1
sigma.alpha	1	1

Multivariate psrf

1

Figure 15: Gelman-Rubin statistics of the missing values imputation

Conclusion & Discussion

Our analyses suggest that familial longevity is more strongly linked to the dynamics of cognitive aging than to baseline cognitive status. At baseline, centenarian offspring differed from controls on several vascular and behavioral risk factors, but once these were adjusted for, TICS01 did not differ meaningfully by group, indicating that parental longevity does not translate into clearly superior initial cognitive performance. However, offspring showed a slower rate of decline than controls, consistent with prior work that links familial longevity to preserved cognition and reduced incidence of cognitive impairment at older ages. The latent class analysis revealed distinct subgroups with stable versus declining trajectories, underscoring substantial heterogeneity in aging patterns and suggesting that resilience to decline may cluster within a subset of individuals, potentially enriched among offspring. The missing-data exploration showed that conclusions about trajectories and group differences are sensitive to how incomplete TICS measures are handled, with mean imputation reversing the apparent direction of change, which is compatible with a Missing Not At Random mechanism. This emphasizes that future work on cognitive aging in centenarian offspring should pair rich longitudinal follow-up with principled missing-data models such as, joint or pattern-mixture approaches to avoid underestimating true decline and mischaracterizing group differences, and should explore biological and environmental pathways (e.g., vascular health, lifestyle, and genetic factors) that might underlie the observed protection.

We would like to note that although the entire project was reviewed and discussed collaboratively as a team, individual sections of the code were completed by different team members. As a result, there may be some inconsistencies in areas such as variable naming or implementation methods. However, all final deliverables and conclusions were agreed upon as a team.

Methods

Difference in variable when comparing offspring of centenarians and control at baseline

Univariate analysis was performed on each baseline characteristic between the control group and the centenarian offspring group to determine potential confounders. For the continuous variables, we checked whether they followed a normal distribution before determining whether to analyze them with a t-test or a Wilcoxon test.

```
hist(CSV.Data$Years.of.Education)
boxplot(Age01 ~ ptype, data = CSV.Data)
qqnorm(CSV.Data$Age01)
```

All variables follow normal distribution (determined by a bell-shaped histogram) except for BMI, which appears right skewed, therefore we conducted a Wilcoxon Test on BMI

```
wilcox.test(BMI~ptype, data = CSV.Data)
t.test(Years.of.Education~ptype, data = CSV.Data)
summary(CSV.Data$BMI)
```

Statistically Significant predictors were noted and utilized in Part 2.

sex (p<0.001) BMI (p=0.017) Smoked (p=0.003) Stroke (p=0.037) Diabetes (p<0.001) HTN (p<0.001) MC.CAD (p<0.001)

```
install.packages("readxl")
library(readxl)
```

```
table <- data
table$ptype <- as.factor(table$ptype)
levels(table$ptype) <- c("Control", "Centennium Offspring")
```

```
install.packages("gtsummary")
```

```

library(gtsummary)
library(dplyr)

vars_to_include <- c("BMI", "Age01", "sex", "SH.Ever.Smoked.", "MC.Aspirin", "MC.Stroke",
  "MC.Diabetes.Mellitus", "MC.HTN", "MC.Coronary.Artery.Disease",
  "MC.Cancer", "MC.Heart.Attack", "Years.of.Education", "TICS01")

tbl_1 <- table %>%
  tbl_summary(
    by = ptype,
    include = vars_to_include,
    type = list(
      BMI ~ "continuous",
      Age01 ~ "continuous",
      Years.of.Education ~ "continuous",
      sex ~ "categorical",
      SH.Ever.Smoked. ~ "categorical",
      MC.Aspirin ~ "categorical",
      MC.Stroke ~ "categorical",
      MC.Diabetes.Mellitus ~ "categorical",
      MC.HTN ~ "categorical",
      MC.Coronary.Artery.Disease ~ "categorical",
      MC.Cancer ~ "categorical",
      MC.Heart.Attack ~ "categorical"
    ),

    statistic = list(
      BMI ~ "{median} ({p25}, {p75})",
      Age01 ~ "{mean} ({sd})",
      Years.of.Education ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    ),

  ) %>%
  add_p(
    test = list(
      BMI ~ "wilcox.test",    # skewed
      Age01 ~ "t.test",      # normal
      Years.of.Education ~ "t.test",
      all_categorical() ~ "chisq.test"
    )
  ) %>%
  modify_header(
    p.value ~ "**p-value**"          # rename p-value column
  )

tbl_1

```

Distributions of TICS score at baseline between offspring of centenarians and controls after adjusting for possible confounders

Model: Normal linear regression in JAGS

$TICS01[i] \sim \text{Normal}(\mu[i], \tau)$

```
mu[i] = beta0 + beta1ptype[i] + beta2Age01[i] + beta3sex[i] + beta4Years.of.Education[i] + beta5*BMI[i] +
...
```

```
model.base <- "
```

```
model{
```

```
  for(i in 1:N){
```

```
    TICS01[i] ~ dnorm(mu[i], tau)
```

Confounder selection from question 1 ($p < 0.05$): sex ($p < 0.001$) BMI ($p = 0.017$) Smoked ($p = 0.003$) Stroke ($p = 0.037$) Diabetes ($p < 0.001$) HTN ($p < 0.001$) MC.CAD ($p < 0.001$)

Age01 is included as standard clinical adjustment

```
  mu[i] <- beta0
```

```
    + beta1 * ptype[i]      # CenOff vs Control (main interest)
```

```
    + beta2 * age[i]        # age at baseline
```

```
    + beta3 * sex[i]        # sex (p<0.001)
```

```
    + beta4 * bmi[i]        # BMI (p=0.037)
```

```
    + beta5 * smoked[i]     # ever smoked (p=0.003)
```

```
    + beta6 * stroke[i]     # stroke history (p=0.037)
```

```
    + beta7 * dm[i]         # diabetes (p=0.001)
```

```
    + beta8 * htn[i]        # hypertension (p<0.001)
```

```
    + beta9 * cad[i]        # coronary artery disease (p=0.001)
```

```
  }
```

Priors: Weak prior is used:

beta_k ~ dnorm(0, 0.001) [mean 0, precision 0.001 = variance 1000] -> very weakly informative, essentially flat

tau ~ dgamma(0.01, 0.01) -> weakly informative for precision

```
  beta0 ~ dnorm(0, 0.001)
```

```
  beta1 ~ dnorm(0, 0.001)
```

```
  beta2 ~ dnorm(0, 0.001)
```

```
  beta3 ~ dnorm(0, 0.001)
```

```
  beta4 ~ dnorm(0, 0.001)
```

```
  beta5 ~ dnorm(0, 0.001)
```

```
  beta6 ~ dnorm(0, 0.001)
```

```
  beta7 ~ dnorm(0, 0.001)
```

```
  beta8 ~ dnorm(0, 0.001)
```

```
  beta9 ~ dnorm(0, 0.001)
```

```
  tau ~ dgamma(0.01, 0.01)
```

```
  sigma <- 1/sqrt(tau)
```

```
}
```

Missing data strategy:

- Outcome (TICS01): 2 missing values -> passed as NA, JAGS handles them
- Covariates: mean imputation before passing to JAGS

```
# Impute covariates (mean imputation, same as Q3)
```

```
Age01_imp <- impute_mean(TICS$Age01)
```

```
BMI_imp <- impute_mean(TICS$BMI)
```

```
ptype_b <- impute_mean(TICS$ptype.num)
```

```
sex_b <- impute_mean(TICS$sex.num)
```

```
smoked_b <- impute_mean(TICS$smoked.num)
```

```
stroke_b <- impute_mean(TICS$stroke.num)
```

```
dm_b      <- impute_mean(TICS$dm.num)
htn_b     <- impute_mean(TICS$htn.num)
cad_b     <- impute_mean(TICS$cad.num)
```

Hierarchical model to investigate differences in the rate of change of TICS between offspring of centenarians and controls

We used a Bayesian hierarchical linear model to investigate whether the rate of change in TICS differed between centenarian offspring and controls. A time variable was created from the age-at-visit variables, and the follow-up TICS scores (TICS02–TICS05) were used as outcomes, with TICS01 included as a baseline covariate and additional adjustment for baseline risk factors (baseline age, BMI, sex, smoking, hypertension, diabetes, stroke, and coronary artery disease). For subject i at visit j , we specified a normal likelihood with a subject-specific random intercept $\alpha[i]$, a common slope for controls, and a group-by-time interaction capturing the offspring–control difference in slope, plus regression terms for the baseline covariates. Random intercepts were modeled with a normal distribution to account for within-person correlation over time. Weakly informative Normal priors were assigned to all regression coefficients and Gamma priors to the precision parameters, and the model was fit in JAGS using four MCMC chains with 3,000 burn-in iterations, 4 MCMC chains, and 30,000 retained iterations per chain. Convergence was assessed using Gelman–Rubin statistics, trace and density plots for key parameters, and autocorrelation plots for the slope and slope-difference parameters.

```
# Model specification with baseline age
model_q3 <- "
model {
  for (i in 1:N) {
    for (j in 1:4) {
      y[i,j] ~ dnorm(mu[i,j], tau)
      mu[i,j] <- alpha[i]
        + (beta + delta * group[i]) * Time[j]
        + gamma * TICS01[i]
        + b_age * (Age01[i] - mean(Age01[]))
        + b_bmi * (BMI[i] - mean(BMI[]))
        + b_sex * Sex[i]
        + b_smoke * Smoking[i]
        + b_htn * HTN[i]
        + b_diab * Diabetes[i]
        + b_stroke * Stroke[i]
        + b_cad * CAD[i]
    }
    alpha[i] ~ dnorm(mu_alpha, tau_alpha)
  }

  mu_alpha ~ dnorm(0, 0.001)
  beta ~ dnorm(0, 0.001) # control slope
  delta ~ dnorm(0, 0.001) # group difference in slope
  gamma ~ dnorm(0, 0.001)

  b_age ~ dnorm(0, 0.001)
  b_bmi ~ dnorm(0, 0.001)
  b_sex ~ dnorm(0, 0.001)
  b_smoke ~ dnorm(0, 0.001)
  b_htn ~ dnorm(0, 0.001)
  b_diab ~ dnorm(0, 0.001)
  b_stroke ~ dnorm(0, 0.001)
  b_cad ~ dnorm(0, 0.001)
```

```

tau      ~ dgamma(1, 1)
tau_alpha ~ dgamma(1, 1)

sigma      <- 1/sqrt(tau)
sigma_alpha <- 1/sqrt(tau_alpha)
}
"

# Complete cases for all confounders
cc_idx_q3 <- which(complete.cases(TICS[, c("TICS01", "Age01", "BMI", "sex",
                                           "SH.Ever.Smoked.", "MC.HTN",
                                           "MC.Diabetes.Mellitus", "MC.Stroke",
                                           "MC.Coronary.Artery.Disease")]))

data_q3 <- list(
  N      = length(cc_idx_q3),
  y      = y[cc_idx_q3, ],
  Time   = Time,
  group  = TICS$group[cc_idx_q3],
  TICS01 = TICS$TICS01[cc_idx_q3],
  Age01  = TICS$Age01[cc_idx_q3],
  BMI    = TICS$BMI[cc_idx_q3],
  Sex    = TICS$sex[cc_idx_q3],
  Smoking = TICS$SH.Ever.Smoked.[cc_idx_q3],
  HTN     = TICS$MC.HTN[cc_idx_q3],
  Diabetes = TICS$MC.Diabetes.Mellitus[cc_idx_q3],
  Stroke  = TICS$MC.Stroke[cc_idx_q3],
  CAD     = TICS$MC.Coronary.Artery.Disease[cc_idx_q3]
)

# Fit model
jags_q3 <- jags.model(textConnection(model_q3),
                      data      = data_q3,
                      n.adapt   = 3000,
                      n.chains  = 4)

update(jags_q3, 3000)

samples_q3 <- coda.samples(
  jags_q3,
  c("mu_alpha", "beta", "delta", "gamma",
    "b_age", "b_bmi", "b_sex", "b_smoke", "b_htn",
    "b_diab", "b_stroke", "b_cad",
    "sigma", "sigma_alpha"),
  n.iter = 30000
)

samples_q3_combined <- as.mcmc(do.call(rbind, samples_q3))

cat("\nHierarchical Model - Rate of Change \n")
summary(samples_q3)
gelman.diag(samples_q3) # Gelman-Rubin stats

```

```
## Trace + density plots
png("q3_trace_density.png", width = 1200, height = 800, res = 150)
par(mar = c(2, 2, 2, 1))
plot(samples_q3[, c("beta", "delta", "gamma")])
dev.off()

## ACF plots
png("q3_acf.png", width = 800, height = 900, res = 150)
par(mfrow = c(3, 1), mar = c(4, 4, 2, 1))
acf(as.numeric(samples_q3_combined[, "beta"]), main = "", ylab = "beta")
acf(as.numeric(samples_q3_combined[, "delta"]), main = "", ylab = "delta")
acf(as.numeric(samples_q3_combined[, "gamma"]), main = "", ylab = "gamma")
dev.off()
```

Subgroups of study participants with significantly different rates of change of TICS

To assess whether there are distinct groups of study participants with different rates of change in cognitive function over time, we fit a Bayesian latent trajectory mixture model in JAGS. The longitudinal outcome consisted of repeated TICS measurements from follow-up visits:

- TICS02
- TICS03
- TICS04
- TICS05

Baseline cognition, TICS01, was included as a covariate rather than as part of the repeated outcome, so that the latent groups would reflect differences in longitudinal trajectories after accounting for initial cognitive status. Time was defined as the average follow-up time between visits, calculated using differences in age across visits.

For subject i at time j , the model was specified as: $Y_{ij} \sim \text{Normal}(\mu_{ij}, \tau)$ $\mu_{ij} = b0_k + g0_k * \text{TICS01}_i + b1_k * \text{Time}_j$ where k represents the latent class, $b0_k$ is the class-specific intercept, $b1_k$ is the class-specific slope, and $g0_k$ is the effect of baseline cognition.

A two-class mixture model was assumed:

```
w[i] ~ Categorical(theta)
with theta ~ Dirichlet(1,1).
```

Weakly informative priors were used: A complete-case analysis was used, retaining only subjects with non-missing values for TICS01–TICS05. The model was estimated using 3 MCMC chains, with 2,000 adaptation iterations, 5,000 burn-in iterations, and 10,000 sampling iterations per chain. Convergence was assessed using trace plots, Geweke diagnostics, and Gelman-Rubin diagnostics.

```
library(rjags)
library(coda)
dat <- read.csv("C:/Users/Administrator/Desktop/26SPRG/BS849/group_project/Data.csv")

# average visit times
Time <- numeric(4)
Time[1] <- mean(dat$Age02 - dat$Age01, na.rm = TRUE)
Time[2] <- Time[1] + mean(dat$Age03 - dat$Age02, na.rm = TRUE)
Time[3] <- Time[2] + mean(dat$Age04 - dat$Age03, na.rm = TRUE)
Time[4] <- Time[3] + mean(dat$Age05 - dat$Age04, na.rm = TRUE)

# keep complete cases for Q4
use <- complete.cases(dat[, c("TICS01", "TICS02", "TICS03", "TICS04", "TICS05")])
```

```

y <- as.matrix(dat[use, c("TICS02","TICS03","TICS04","TICS05")])
x0 <- dat$TICS01[use]

n.subj <- nrow(y)
par(mfrow = c(1,1),
    mar = c(3,3,1.5,0.5),
    mgp = c(1.6,0.5,0),
    cex.lab = 0.9,
    cex.axis = 0.8,
    tck = -0.02)

plot(c(min(Time), max(Time)), range(y),
     type = "n", xlab = "Time", ylab = "TICS")

for(i in 1:n.subj){
  lines(Time, y[i, ], col = rgb(0,0,0,0.2))
}
model_string <- "
model {
  for(i in 1:n.subj){
    w[i] ~ dcat(theta[])

    for(j in 1:4){
      y[i,j] ~ dnorm(mu[i,j], tau)
      mu[i,j] <- b0[w[i]] + g0[w[i]] * x0[i] + b1[w[i]] * Time[j]
    }
  }

  theta[1:2] ~ ddirch(alpha[])

  for(k in 1:2){
    b1[k] ~ dnorm(0, 0.01)
    g0[k] ~ dnorm(0, 0.001)
    alpha[k] <- 1
  }

  b0[1] ~ dnorm(0, 0.001)
  b0[2] <- b0[1] + delta
  delta ~ dgamma(1, 1)

  tau ~ dgamma(1,1)
}
"
jags_data <- list(
  y = y,
  x0 = x0,
  Time = Time,
  n.subj = n.subj
)

set.seed(123)

mod <- jags.model(

```

```

    textConnection(model_string),
    data = jags_data,
    n.chains = 3,
    n.adapt = 2000
  )

update(mod, 5000)

samp <- coda.samples(
  mod,
  variable.names = c("b0", "b1", "g0", "theta", "tau", "w"),
  n.iter = 10000
)
summary(samp[, c("b0[1]", "b0[2]", "b1[1]", "b1[2]", "g0[1]", "g0[2]", "theta[1]", "theta[2]", "tau")])
plot(samp[, c("b0[1]", "b0[2]", "b1[1]", "b1[2]", "theta[1]", "theta[2]")])
mat <- as.matrix(samp)

w_cols <- grep("^w\\\[", colnames(mat))
w_post <- mat[, w_cols]

group <- apply(w_post, 2, median)

table(group)
prop.table(table(group))
par(mfrow = c(1,1),
    mar = c(3,3,1.5,0.5),
    mgp = c(1.6,0.5,0),
    cex.lab = 0.9,
    cex.axis = 0.8,
    tck = -0.02)

plot(c(min(Time), max(Time)), range(y),
     type = "n", xlab = "Time", ylab = "TICS")

cols <- ifelse(group == 1, rgb(0,0,1,0.2), rgb(1,0,0,0.2))

for(i in 1:n.subj){
  lines(Time, y[i, ], col = cols[i])
}

mean1 <- colMeans(y[group == 1, , drop = FALSE])
mean2 <- colMeans(y[group == 2, , drop = FALSE])

lines(Time, mean1, col = "blue", lwd = 3)
lines(Time, mean2, col = "red", lwd = 3)

legend("topright",
      legend = c("Class 1", "Class 2"),
      col = c("blue", "red"),
      lty = 1, lwd = 3, bty = "n")
prob_class1 <- colMeans(w_post == 1)
prob_class2 <- colMeans(w_post == 2)

head(data.frame(group, prob_class1, prob_class2))

```

```

# Check convergence
params_conv1 <- c("b0[1]", "b0[2]", "b1[1]", "b1[2]")
params_conv2 <- c("g0[1]", "g0[2]", "theta[1]", "tau")

geweke.diag(samp[, params_conv1], frac1 = 0.1, frac2 = 0.5)
geweke.diag(samp[, params_conv2], frac1 = 0.1, frac2 = 0.5)

gelman.diag(samp[, params_conv1], multivariate = FALSE)
gelman.diag(samp[, params_conv2], multivariate = FALSE)

par(mar = c(2,2,1,1))
gelman.plot(samp[, params_conv1])

par(mar = c(2,2,1,1))
gelman.plot(samp[, params_conv2])

```

Robustness of the results to the missing data mechanism

Initial analysis of how much of the data was missing using the script below. Then the association between the covariates and NA values was conducted using a logistic regression model to determine if the data is not missing completely at random. `{r}` `miss_summary <- colSums(is.na(TICS[, c("TICS01", "TICS02", "TICS03", "TICS04", "TICS05")]))` `print(round(miss_summary / N * 100, 1))`

```

miss_summary <- colSums(is.na(TICS[, c("TICS01", "TICS02", "TICS03", "TICS04", "TICS05")]))
print(round(miss_summary / N * 100, 1))

```

```

TICS$miss_TICS05 <- as.numeric(is.na(TICS$TICS05)) # binary missingness indicator
model_mar <- glm(miss_TICS05 ~ group + TICS01 + Age01 + sex,
  data = TICS,
  family = binomial)

```

```

TICS$miss_TICS05 <- as.numeric(is.na(TICS$TICS05)) # binary missingness indicator
model_mar <- glm(miss_TICS05 ~ group + TICS01 + Age01 + sex, data = TICS, family = binomial)

```

Once a significant association was established, the model from part 3 was reused with imputed data. To impute the data, first, the missing age values were replaced with the average of non missing values from that corresponding visit.

```

for (col in c("Age02", "Age03", "Age04", "Age05")) {
  TICS_imp[[col]][is.na(TICS_imp[[col]])] <- mean(TICS_imp[[col]], na.rm = TRUE)
}

```

Similarly, the TICS02-TICS05 NA values were imputed the average of the scores that were not missing.

```

for (col in c("TICS02", "TICS03", "TICS04", "TICS05")) {
  TICS_imp[[col]][is.na(TICS_imp[[col]])] <- mean(TICS_imp[[col]], na.rm = TRUE)
}

```

Lastly, there were some missing BMI values, which were fixed with the average BMI values of the participants.

```

TICS_imp$BMI[is.na(TICS_imp$BMI)] <- mean(TICS_imp$BMI, na.rm = TRUE)

```